Francis Omoniyi Emmanuel

Ph.D. Candidate in Applied Mathematics
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Ph.D. candidate in Applied Mathematics with research interests at the intersection of mathematical biology, dynamical systems, survival analysis, longitudinal health data, and scientific computing. My dissertation integrates longitudinal survival analysis with a physical activity-stratified compartmental model of cardiovascular disease and mortality. I have substantial undergraduate teaching experience from developmental algebra through Calculus I and emphasize conceptual understanding, active student participation, clear mathematical communication, and connections between mathematics and real-world applications.

Research Interests

Mathematical Biology • Dynamical Systems • Differential Equations • Survival Analysis • Longitudinal Data Analysis • Scientific Computing • Data-Informed Mathematical Modeling • Parameter Estimation and Inverse Problems

Education

Aug. 2022 – Expected Apr. 2027
Ph.D. in Applied Mathematics
University of Alabama at Birmingham, Birmingham, Alabama
Dissertation: *Mathematical Modeling of Cardiovascular Disease–Mortality Dynamics: A Data-Informed Framework with Physical Activity Effect Modification*

Aug. 2022 – Apr. 2024
Master of Science in Mathematics, Non-Thesis
University of Alabama at Birmingham, Birmingham, Alabama

Oct. 2017 – Aug. 2021
Bachelor of Science in Mathematics
University of Ilorin, Ilorin, Nigeria
Minor: Statistics
First Class Honors

Research Experience

Aug. 2022 – Present
Graduate Researcher
Department of Mathematics, University of Alabama at Birmingham

- Develop and analyze differential-equation models for chronic and infectious disease processes, with emphasis on mathematical biology and population health.
- Construct longitudinal counting-process datasets and analyze time-to-event outcomes using Cox proportional hazards and extended Cox models with time-varying covariates.
- Investigate nonproportional hazards, interaction effects, mortality risk, and incident cardiovascular disease using SAS and R.
- Develop a physical activity-stratified compartmental model linking cardiovascular disease, behavior, and mortality, with empirically informed time-dependent transition and mortality mechanisms.
- Study mathematical properties of the compartmental system, including existence and uniqueness, nonnegativity, boundedness, and numerical behavior.
- Implement numerical simulations and parameter calibration using computational methods including the classical fourth-order Runge–Kutta method.

Jan. – Jul. 2022
Data Analyst Intern
Health Strategy & Delivery Foundation, Nigeria

- Analyzed more than 10,000 patient records from a diabetes awareness and care project.
- Cleaned, validated, transformed, and organized health datasets using Python and Microsoft Excel.
- Identified data-quality issues and summarized quantitative trends for project stakeholders.

Teaching Experience

Fall 2026 — **Instructor of Record**
University of Alabama at Birmingham, Birmingham, Alabama

- Calculus I (MA 125)
- Develop instructional materials and assessments, evaluate student work, and promote conceptual understanding and independent problem solving.

Aug. 2022 – Present — **Graduate Teaching Assistant**
University of Alabama at Birmingham, Birmingham, Alabama

- Teach and support undergraduate mathematics courses ranging from developmental algebra through Calculus I.
- Prepare instructional materials, quizzes, examinations, review materials, and other course content.
- Hold office and tutoring hours and provide individualized academic support to students with varied levels of preparation.
- Use Canvas, ALEKS, MyLab Math, iClicker, Poll Everywhere, Desmos, and graphing technology to support instruction and formative assessment.

Summer 2026 — **Upward Bound Mathematics Instructor**
University of Alabama at Birmingham

- Designed and taught Algebra II with Trigonometry and Precalculus for high-school students preparing for college mathematics.
- Developed lessons, guided-practice activities, assessments, and individualized academic support.

Jan. 2020 – Dec. 2021 — **Mathematics Lecturer**
Peniel Advanced Pre-Versity, Nigeria

- Taught Advanced Mathematics I and II.
- Developed instructional materials, assessments, examination-review materials, and academic support activities.

Courses Taught or Supported

- **MA 125 – Calculus I:** Fall 2022, Spring 2023, Fall 2026
- **MA 107 – Precalculus Algebra/Trigonometry:** Fall 2025, Spring 2026
- **MA 106 – Precalculus Trigonometry:** Summer 2025
- **MA 102 – Intermediate Algebra:** Spring 2025
- **MA 105 – Precalculus Algebra:** Spring, Summer, and Fall 2024
- **MA 098 – Basic Algebra:** Fall 2023
- **Algebra II with Trigonometry and Precalculus:** UAB Upward Bound, Summer 2026
- **MAT 111/112 – Advanced Mathematics I and II:** 2020–2021

Peer-Reviewed Publications

1. O. Odetunde, T. Alesinloye, and **O. Francis**. “Modeling the Dynamics of Type 2 Diabetes with Modifiable Lifestyle Influence.” *Journal of the Nigerian Society of Physical Sciences*, 8, Article 3161, 2026. doi:10.46481/jnsps.2026.3161.

2. O. T. Adedeji, T. S. Alesinloye, and **E. O. Francis**. “A Social Influence-Driven Compartmental Model of Obesity Dynamics with Multi-Class Severity and Relapse.” *Franklin Open*, 16, Article 100698, 2026. doi:10.1016/j.fraope.2026.100698.

3. O. Odetunde, **O. Francis**, and T. Alesinloye. “Optimal Control Analysis of Hepatitis B Transmission Dynamics.” *World Journal of Advanced Research and Reviews*, 24, 2024. doi:10.30574/wjarr.2024.24.2.3532.

4. B. Teri, **E. O. Francis**, et al. “The Role of Big Data in Driving Advancements in Deep Learning: Opportunities, Challenges, and Future Directions.” *World Journal of Advanced Research and Reviews*, 24, 690–696, 2024. doi:10.30574/wjarr.2024.24.2.3390.

Papers in Preparation

1. **E. O. Francis**. "Physical Activity, Cardiovascular Disease, and Mortality: A Longitudinal and Mathematical Modeling Approach."
2. **E. O. Francis**. "On the Dynamics of HIV–Cancer Comorbidity with Physical Activity-Dependent Mortality."

Selected Research Projects

Cardiovascular Disease, Physical Activity, and Mortality

- Analyze longitudinal Health and Retirement Study data using Cox and extended Cox models with time-varying covariates.
- Develop a physical activity-stratified compartmental model with non-CVD and CVD states and mortality mechanisms informed by survival analysis.

Obesity Dynamics

- Contributed to a severity-stratified compartmental model incorporating social influence, recovery, multiple obesity classes, relapse, mathematical analysis, and numerical simulation.

Type 2 Diabetes Modeling

- Studied a mathematical model of Type 2 diabetes incorporating modifiable lifestyle, genetic, and environmental influences.

Hepatitis B Transmission and Optimal Control

- Investigated disease-transmission dynamics and intervention strategies using compartmental differential equations, numerical simulation, and optimal control.

Research Presentations

Jan. 2027 **"Integrating Longitudinal Survival Analysis with a Physical Activity-Stratified Compartmental Model of Cardiovascular Disease and Mortality."**
Accepted contributed presentation, AMS Contributed Paper Session on Mathematical Biology, II, Joint Mathematics Meetings, Chicago, Illinois.

Apr. 2026 **"Time-Dependent Effects of Leisure-Time Physical Activity on Mortality Among U.S. Adults: A Survey-Weighted Cox Analysis."**
Biostatistics Research Presentation, University of Alabama at Birmingham.

Mar. 2026 **"Optimal Control Analysis Applied to the Transmission Dynamics of Hepatitis B to Primary Liver Cancer."**
Department of Mathematics Seminar, University of Alabama at Birmingham.

Selected Conferences & Workshops

Jul. 2026 **International Congress of Mathematicians (ICM 2026)**
Graduate student participant.

2025 **IMSI Opening Tutorial: Mathematical and Statistical Foundations of Data Assimilation for Digital Twins**
Funded participant.

Honors, Awards & Funded Academic Participation

2026 **James R. Ward, Sr. Memorial Endowed Award**
UAB Department of Mathematics

2026 **International Congress of Mathematicians Graduate Student Travel Grant**
American Mathematical Society

2025 **Outstanding Graduate Teaching Assistant Award**
University of Alabama at Birmingham

2025 **Funded Participant**
IMSI Opening Tutorial on Data Assimilation for Digital Twins

2021 **Best Tutor of the Year**
Faculty of Physical Sciences, University of Ilorin

Academic Service & Outreach

2026 **Volunteer Mathematics Outreach Presenter**
"Spill the Mathematical Tea Party," UAB Department of Mathematics

2026 **Proctor**
Alabama Statewide High School Mathematics Contest, University of Alabama at Birmingham

2025 **Proctor**
UAB Math Talent Search

2024 **Judge and Grader**
UAB Middle School Mathematics Tournament

Technical Skills

Mathematical Modeling	Ordinary differential equations, compartmental models, dynamical systems, qualitative analysis, parameter estimation, numerical simulation
Statistical Methods	Survival analysis, Cox proportional hazards models, extended Cox models, longitudinal data analysis, time-varying covariates, proportional-hazards diagnostics
Numerical Methods	Runge–Kutta methods, numerical simulation, sensitivity analysis, parameter calibration, visualization
Software	Python, R, SAS, Maple, Microsoft Excel, LaTeX, Overleaf
Teaching Technology	Canvas, ALEKS, MyLab Math, Pearson, iClicker, Poll Everywhere, Desmos

Professional Development

2026 **Lean Into Future Learning (LIFT) Teaching Certificate**
UAB Center for Teaching and Learning
Training in course design, evidence-based pedagogy, student engagement, inclusive teaching, assessment, and reflective practice.

2025 **Data-Driven Problem Solving and Neural Computing: From Prediction to Prescription, No-Code to High-Code**
National Institute of Statistical Sciences

Professional Memberships

- American Mathematical Society (AMS)
- Society for Industrial and Applied Mathematics (SIAM)